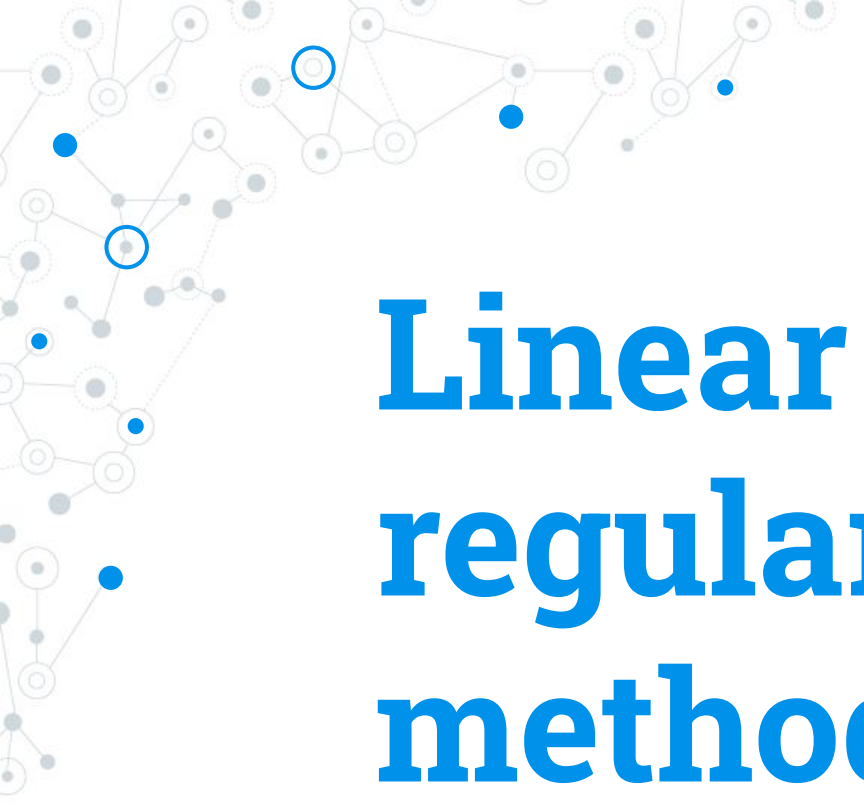
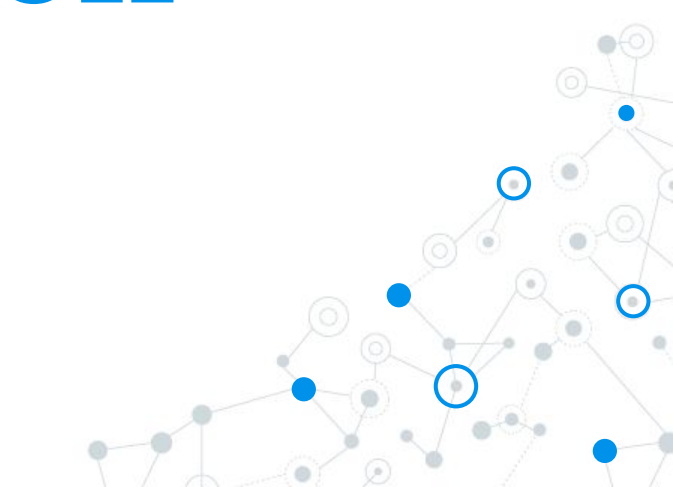




# Linear regularization methods

Rodrigo Gonzalez, PhD



# Hello!

## I am Rodrigo Gonzalez, PhD

You can find me at

[rodrigo.gonzalez@ingenieria.uncuyo.edu.ar](mailto:rodrigo.gonzalez@ingenieria.uncuyo.edu.ar)





1.

# Linear regression

A review

## Linear regression

$$Y = \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p + \epsilon \quad (6.1)$$

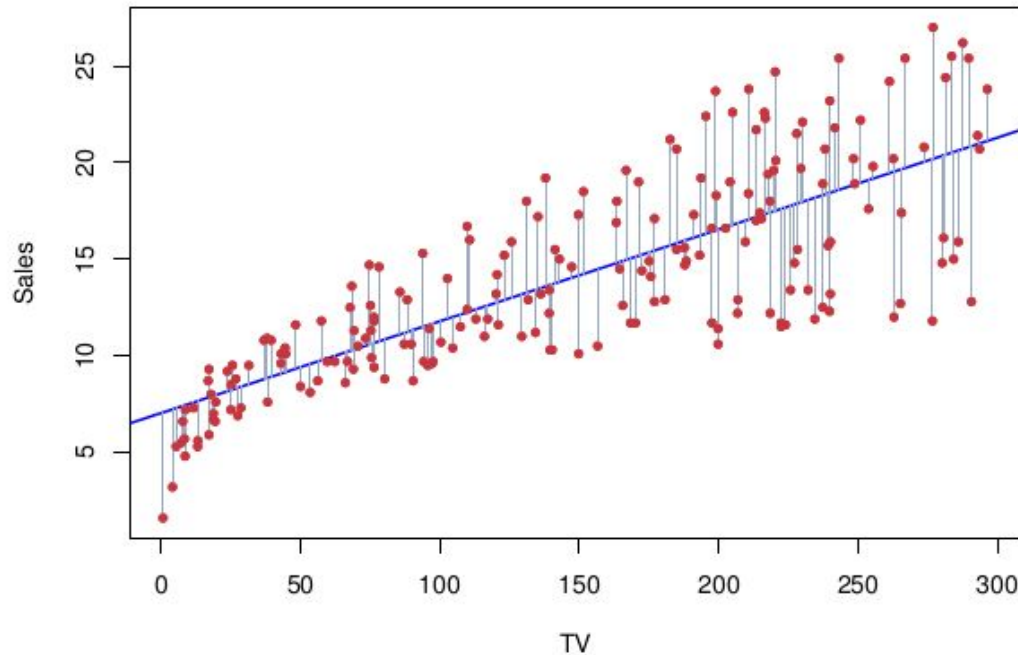
Commonly used to describe the relationship between a response  $Y$  and a set of variables  $X_1, X_2, \dots, X_p$ .

The least squares fitting procedure estimates  $\beta_0, \beta_1, \dots, \beta_p$  using the values that minimize

$$\text{RSS} = \sum_{i=1}^n \left( y_i - \beta_0 - \sum_{j=1}^p \beta_j x_{ij} \right)^2.$$

$$\text{RSS} = e_1^2 + e_2^2 + \cdots + e_n^2,$$

## Linear regression



**FIGURE 3.1.** For the **Advertising** data, the least squares fit for the regression of **sales** onto **TV** is shown. The fit is found by minimizing the residual sum of squares. Each grey line segment represents a residual. In this case a linear fit captures the essence of the relationship, although it overestimates the trend in the left of the plot.

# Linear regression

How to evaluate a linear regression model?

- ◎ R2 statistic

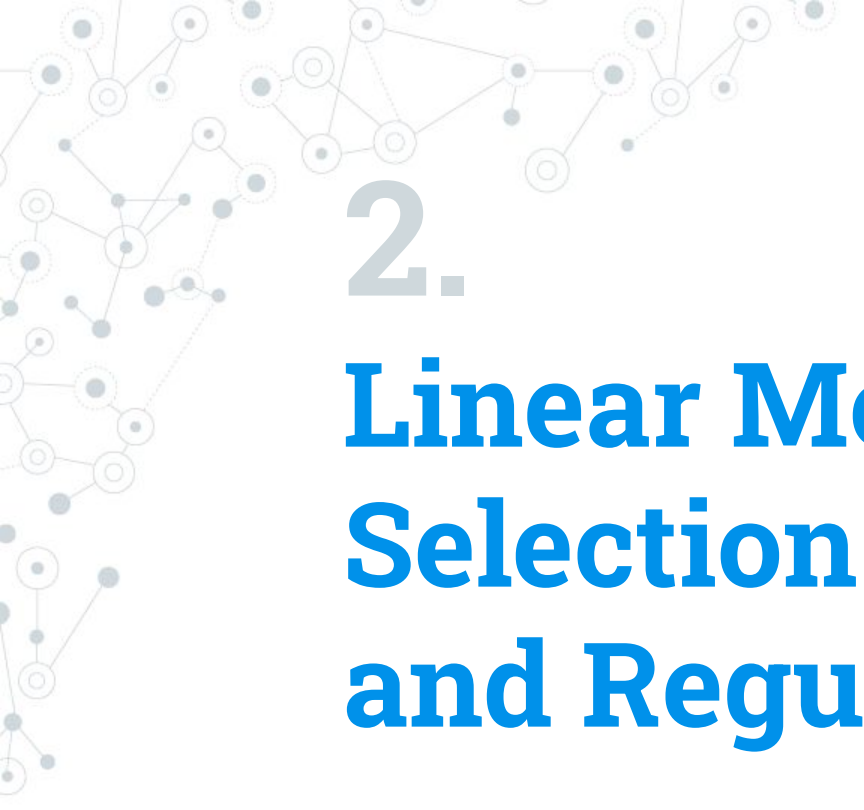
$$R^2 = \frac{TSS - RSS}{TSS} = 1 - \frac{RSS}{TSS} \quad TSS = \sum (y_i - \bar{y})^2$$

- R2 compares the LR performance vs the mean performance.
- The mean is your baseline solution!

- ◎ P-value

- Is there no relationship between X and Y? (null hypothesis).
- A small p-value indicates that it is unlikely to observe such a substantial association between the predictor and the response due to chance.

MAE on testing data.



2.

# Linear Model Selection and Regularization

Ridge and Lasso

# Least squares fitting problems

Why might we want to use another fitting procedure instead of least squares?

- ◎ Prediction Accuracy
  - If  $n$  (the number of observations) is not much larger than  $p$  (the number of variables or predictor or features), then there can be a lot of variability in the least squares fit. Sometimes we need to shrink  $p$ .
- ◎ Model Interpretability
  - Including such irrelevant variables leads to unnecessary complexity in the resulting model. We see some approaches for automatically performing *feature selection* or variable selection.



# Alternatives methods to linear regression

## ◎ Subset Selection

- Identify a subset of the predictors that we believe to be related to the response.
- $2^p$  possible models.
- What if you have 30 predictors?

## ◎ Shrinkage

- This approach involves fitting a model involving all  $p$  predictors. However, the estimated coefficients are shrunk towards zero relative to the least squares estimates.
- Ridge regression and The Lasso



3.

# Ridge regression

## Ridge regression

$$Y = \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p + \epsilon \quad (6.1)$$

Linear regression fitting

$$\text{RSS} = \sum_{i=1}^n \left( y_i - \beta_0 - \sum_{j=1}^p \beta_j x_{ij} \right)^2 .$$

Ridge regression fitting

$$\sum_{i=1}^n \left( y_i - \beta_0 - \sum_{j=1}^p \beta_j x_{ij} \right)^2 + \lambda \sum_{j=1}^p \beta_j^2 = \text{RSS} + \lambda \sum_{j=1}^p \beta_j^2, \quad (6.5)$$

$\lambda \geq 0$  is a tuning parameter

## Tuning parameter

$$Y = \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p + \epsilon \quad (6.1)$$

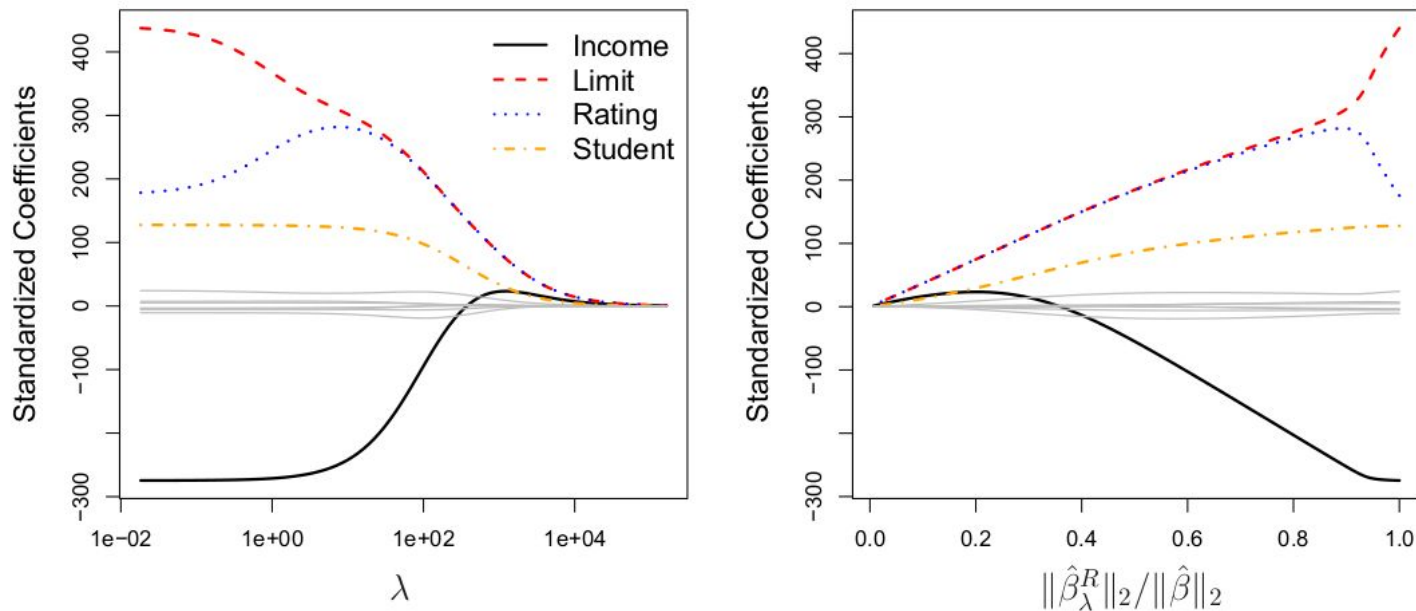
$$\sum_{i=1}^n \left( y_i - \beta_0 - \sum_{j=1}^p \beta_j x_{ij} \right)^2 + \lambda \sum_{j=1}^p \beta_j^2 = \text{RSS} + \lambda \sum_{j=1}^p \beta_j^2, \quad (6.5)$$

The second term is the *shrinkage penalty*. Not to the intercept  $\beta_0$ .

$\lambda \geq 0$  is a tuning parameter

- ⊙  $\lambda = 0$ , the penalty term has no effect.
- ⊙ However, as  $\lambda \rightarrow \infty$ , the impact of the shrinkage penalty grows, and the ridge regression coefficient estimates will approach zero.
- ⊙ Selecting a good value for  $\lambda$  is critical.

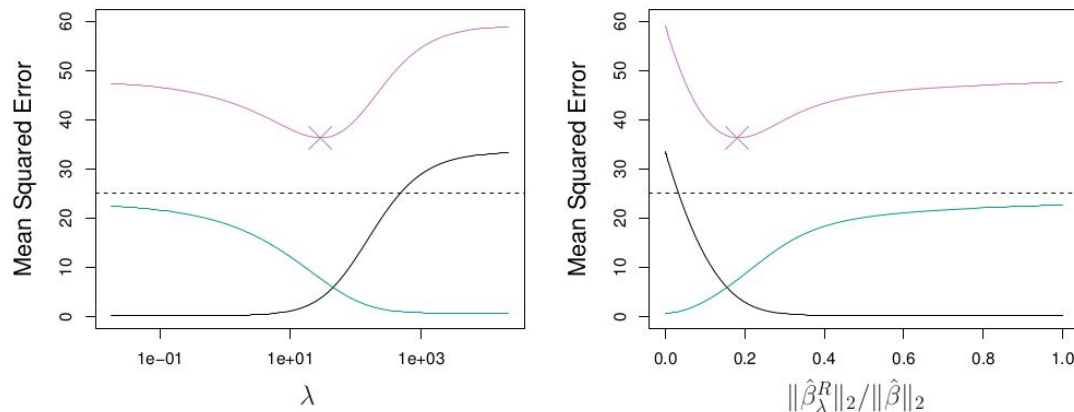
## Tuning parameter



**FIGURE 6.4.** The standardized ridge regression coefficients are displayed for the **Credit** data set, as a function of  $\lambda$  and  $\|\hat{\beta}_\lambda^R\|_2 / \|\hat{\beta}\|_2$ .

## Ridge regression advantage over least squares

Bias-variance trade-off 
$$E \left( y_0 - \hat{f}(x_0) \right)^2 = \text{Var}(\hat{f}(x_0)) + [\text{Bias}(\hat{f}(x_0))]^2 + \text{Var}(\epsilon). \quad (2.7)$$



**FIGURE 6.5.** Squared bias (black), variance (green), and test mean squared error (purple) for the ridge regression predictions on a simulated data set, as a function of  $\lambda$  and  $\|\hat{\beta}_\lambda^R\|_2 / \|\hat{\beta}\|_2$ . The horizontal dashed lines indicate the minimum possible MSE. The purple crosses indicate the ridge regression models for which the MSE is smallest.

$p = 45$  predictors and  $n = 50$  observations, The minimum MSE is achieved at approximately  $\lambda = 30$



# 4. **The Lasso**

## The Lasso

Ridge regression always includes the  $p$  predictors.

The lasso forces some of the coefficient estimates to be exactly equal to zero.

$$Y = \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p + \epsilon \quad (6.1)$$

$$\sum_{i=1}^n \left( y_i - \beta_0 - \sum_{j=1}^p \beta_j x_{ij} \right)^2 + \lambda \sum_{j=1}^p |\beta_j| = \text{RSS} + \lambda \sum_{j=1}^p |\beta_j|. \quad (6.7)$$

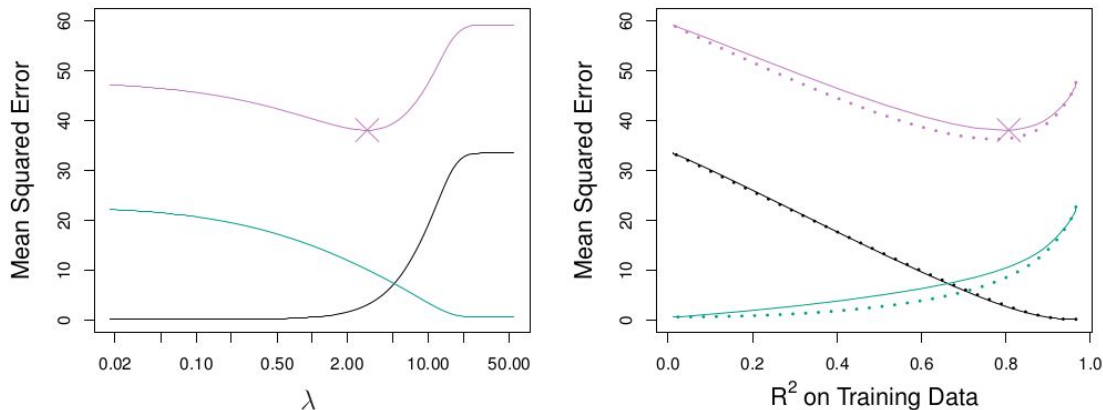
$\lambda \geq 0$  is a tuning parameter.

The lasso performs feature selection.



# The Lasso vs Ridge

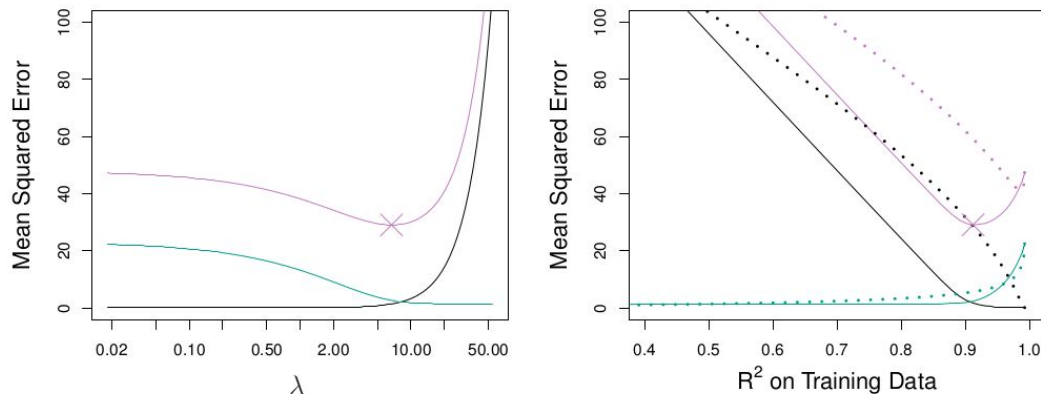
In this example, all 45 predictors are related to the response.



**FIGURE 6.8.** Left: Plots of squared bias (black), variance (green), and test MSE (purple) for the lasso on a simulated data set. Right: Comparison of squared bias, variance, and test MSE between lasso (solid) and ridge (dotted). Both are plotted against their  $R^2$  on the training data, as a common form of indexing. The crosses in both plots indicate the lasso model for which the MSE is smallest.

# The Lasso vs Ridge

In this example, only 2 predictors are related to the response.



**FIGURE 6.9.** Left: Plots of squared bias (black), variance (green), and test MSE (purple) for the lasso. The simulated data is similar to that in Figure 6.8, except that now only two predictors are related to the response. Right: Comparison of squared bias, variance, and test MSE between lasso (solid) and ridge (dotted). Both are plotted against their  $R^2$  on the training data, as a common form of indexing. The crosses in both plots indicate the lasso model for which the MSE is smallest.

A decorative network diagram in the top-left corner, featuring a complex web of interconnected nodes and lines, with some nodes highlighted in blue and others in grey.

5.

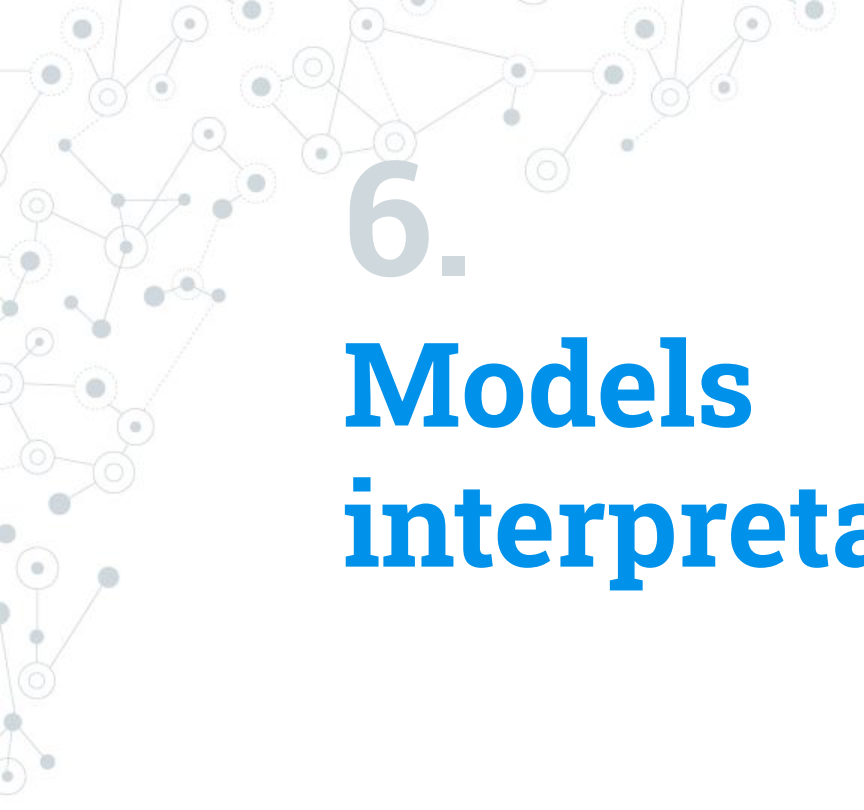
# Data standardization

## Data standardization

It is best to apply ridge regression or lasso regression after standardizing the predictors, using the formula:

$$\tilde{x}_{ij} = \frac{x_{ij}}{\sqrt{\frac{1}{n} \sum_{i=1}^n (x_{ij} - \bar{x}_j)^2}}, \quad (6.6)$$

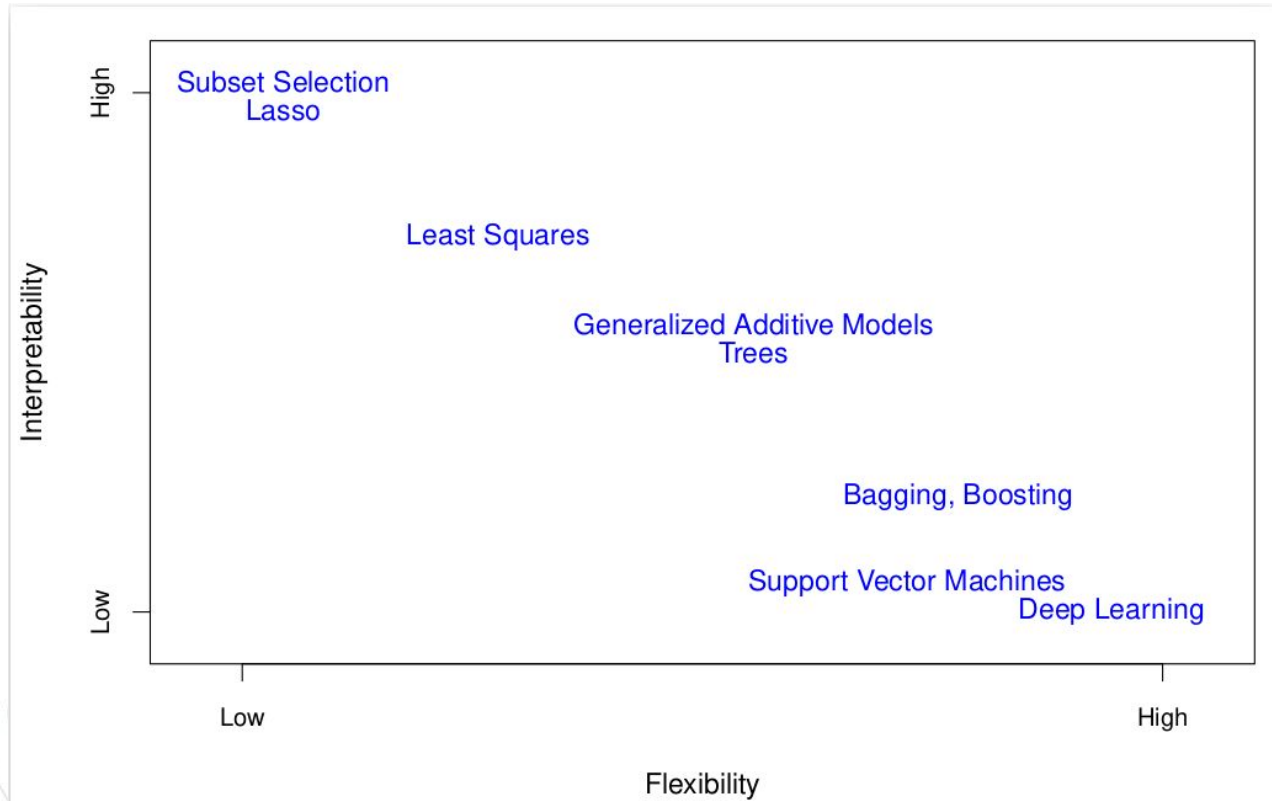
so that they are all *on the same scale*. In (6.6), the denominator is the estimated standard deviation of the  $j$ th predictor. Consequently, all of the standardized predictors will have a standard deviation of one. As a result the final fit will not depend on the scale on which the predictors are measured.



6.

# Models interpretability

# Tradeoff between flexibility and interpretability

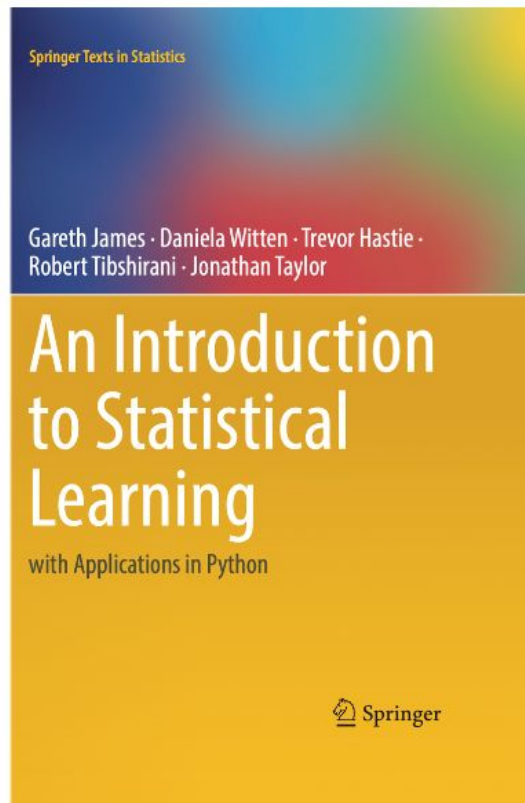


A decorative network diagram in the top-left corner, featuring a complex web of interconnected nodes and lines. The nodes are represented by small circles, some of which are larger and have concentric circles, suggesting a hierarchical or multi-layered structure. The lines are thin and gray, connecting the nodes in a non-linear fashion.

# 7.

## **The book**

# An Introduction to Statistical Learning With Application in Python



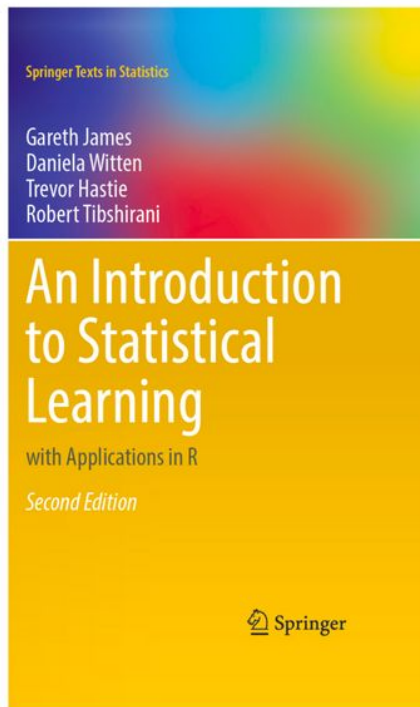
◎ <https://www.statlearning.com/>

## Chapter 6

◎ [https://hastie.su.domains/ISLP/ISLP\\_website.pdf](https://hastie.su.domains/ISLP/ISLP_website.pdf)



# An Introduction to Statistical Learning With Application in R



© <https://www.statlearning.com/>

© [https://hastie.su.domains/ISLR2/ISLRv2\\_website.pdf](https://hastie.su.domains/ISLR2/ISLRv2_website.pdf)



# Thanks!

## Any questions?

# Instructions for use

## EDIT IN GOOGLE SLIDES

Click on the button under the presentation preview that says "Use as Google Slides Theme".

You will get a copy of this document on your Google Drive and will be able to edit, add or delete slides.

You have to be signed in to your Google account.

## EDIT IN POWERPOINT®

Click on the button under the presentation preview that says "Download as PowerPoint template". You will get a .pptx file that you can edit in PowerPoint.

Remember to download and install the fonts used in this presentation (you'll find the links to the font files needed in the [Presentation design slide](#))

**More info on how to use this template at**  
**[www.slidescarnival.com/help-use-presentation-template](http://www.slidescarnival.com/help-use-presentation-template)**

This template is free to use under [Creative Commons Attribution license](#). You can keep the Credits slide or mention SlidesCarnival and other resources used in a slide footer.

A decorative graphic at the top of the slide featuring a network of interconnected nodes and lines, resembling a molecular or digital structure. A central node is highlighted with a blue double quote symbol.

“

*Quotations are commonly printed  
as a **means of inspiration** and to  
invoke philosophical thoughts  
from the reader.*



## This is a slide title

- ◎ Here you have a list of items
- ◎ And some text
- ◎ But remember not to overload your slides with content

Your audience will listen to you or read the content, but won't do both.



# Big concept

Bring the attention of your audience over a key concept using icons or illustrations



## You can also split your content

### **White**

Is the color of milk and fresh snow, the color produced by the combination of all the colors of the visible spectrum.

### **Black**

Is the color of ebony and of outer space. It has been the symbolic color of elegance, solemnity and authority.

## In two or three columns

### **Yellow**

Is the color of gold, butter and ripe lemons. In the spectrum of visible light, yellow is found between green and orange.

### **Blue**

Is the colour of the clear sky and the deep sea. It is located between violet and green on the optical spectrum.

### **Red**

Is the color of blood, and because of this it has historically been associated with sacrifice, danger and courage.



## A picture is worth a thousand words

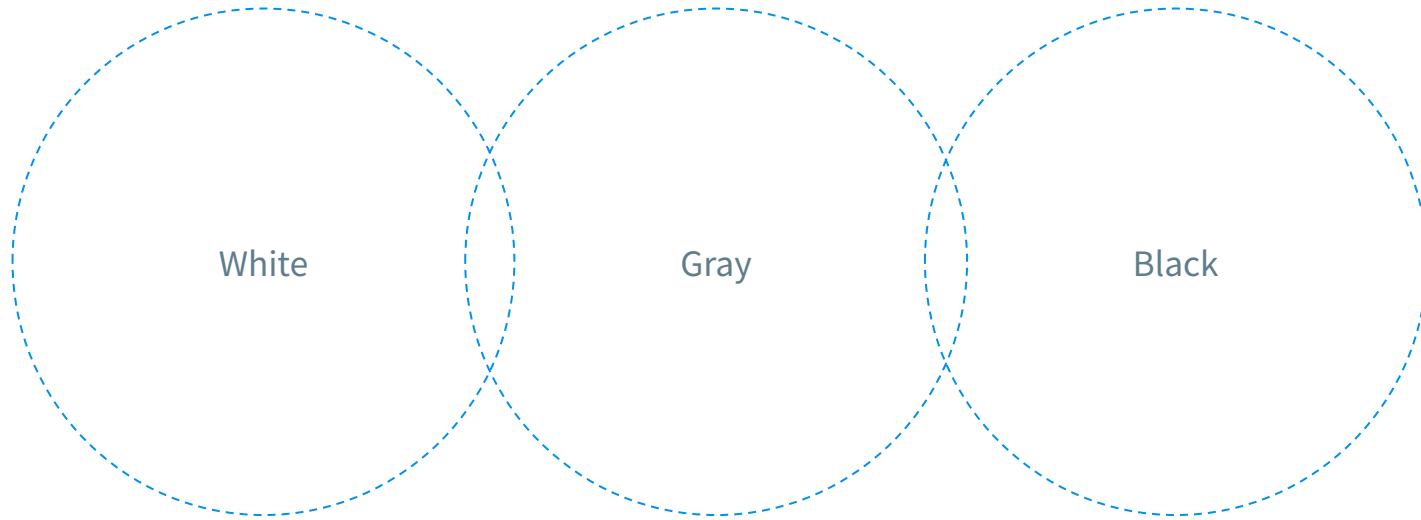
A complex idea can be conveyed with just a single still image, namely making it possible to absorb large amounts of data quickly.



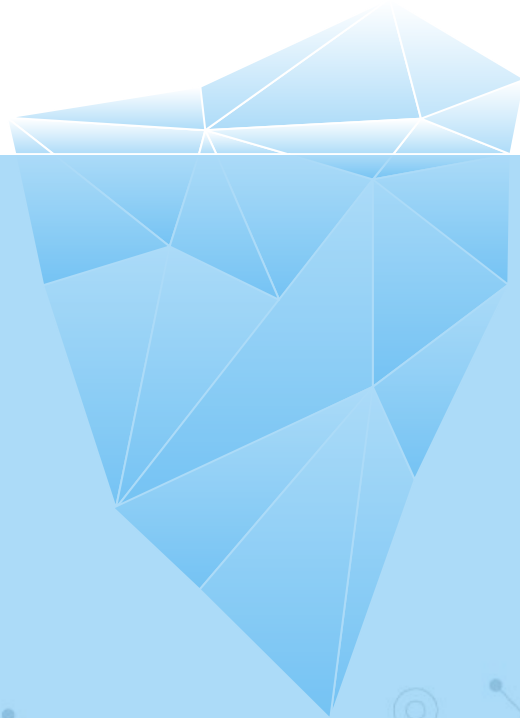


**Want big  
impact?**  
Use big image.

## Use charts to explain your ideas



## Or diagrams to explain complex ideas



### Example text.

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Nam venenatis nisi at nisl tempor, et luctus diam lobortis. Nulla sit amet metus consequat velit iaculis tempor.

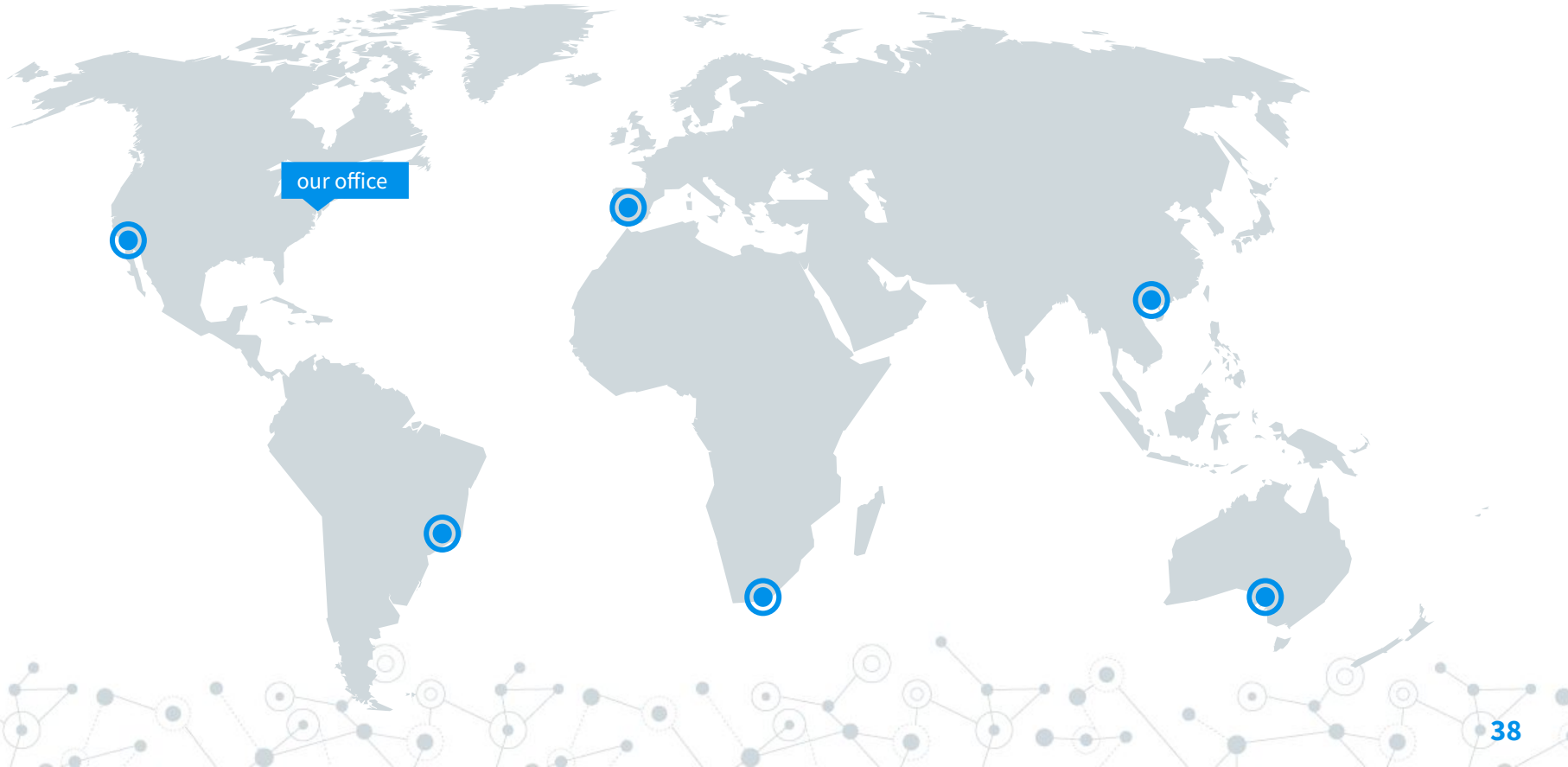
### Example text.

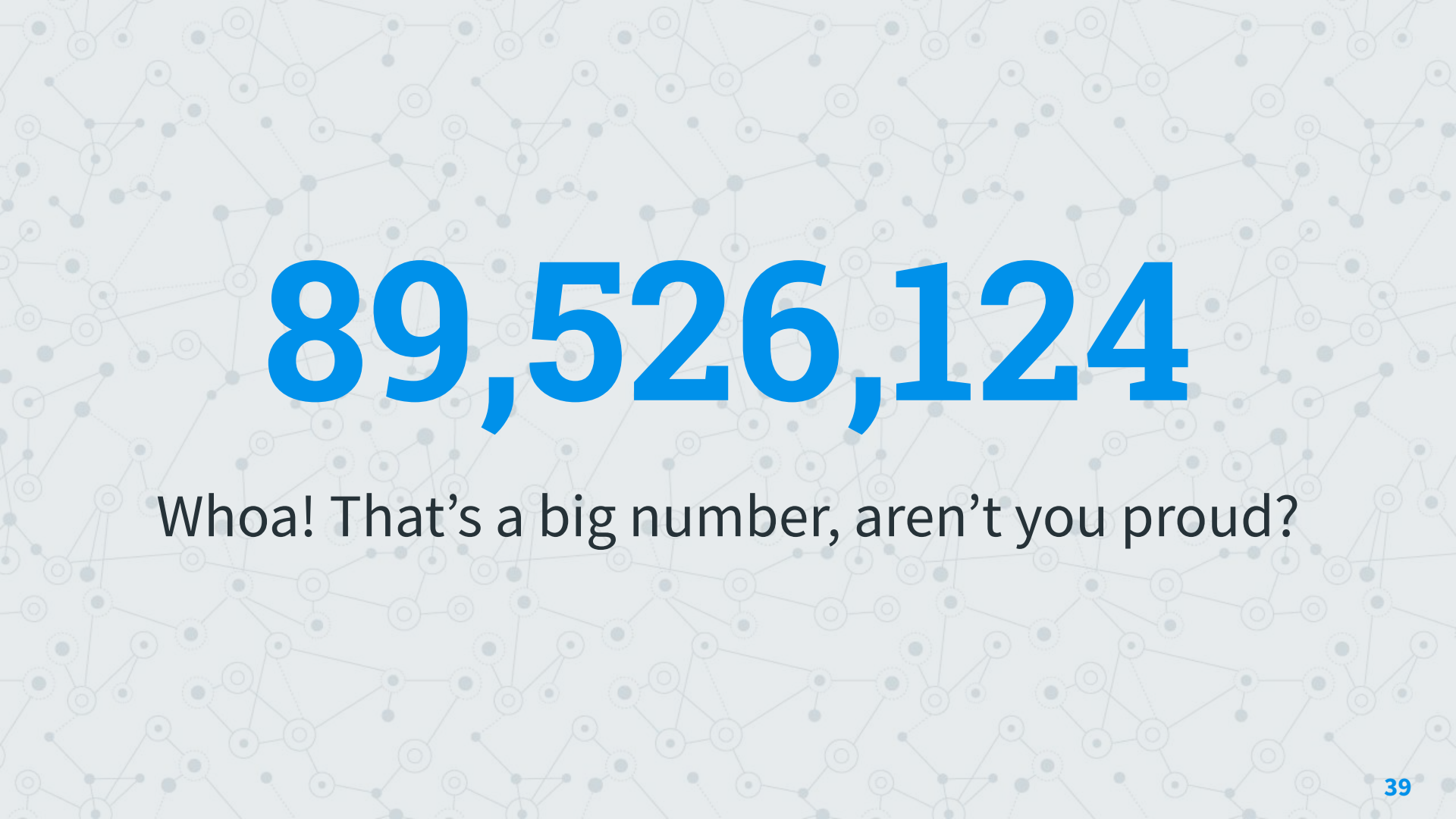
Lorem ipsum dolor sit amet, consectetur adipiscing elit. Nam venenatis nisi at nisl tempor, et luctus diam lobortis. Nulla sit amet metus consequat velit iaculis tempor.

## And tables to compare data

|        | A         | B         | C         |
|--------|-----------|-----------|-----------|
| Yellow | <b>10</b> | <b>20</b> | <b>7</b>  |
| Blue   | <b>30</b> | <b>15</b> | <b>10</b> |
| Orange | <b>5</b>  | <b>24</b> | <b>16</b> |

# Maps



The background of the slide features a light gray network pattern. It consists of numerous small circles, some of which are double-lined, connected by thin, light gray lines. These connections form a complex, web-like structure that fills the entire background.

# 89,526,124

Whoa! That's a big number, aren't you proud?

# Presentation design

This presentations uses the following typographies and colors:

- Titles: **Roboto Slab**
- Body copy: **Source Sans Pro**

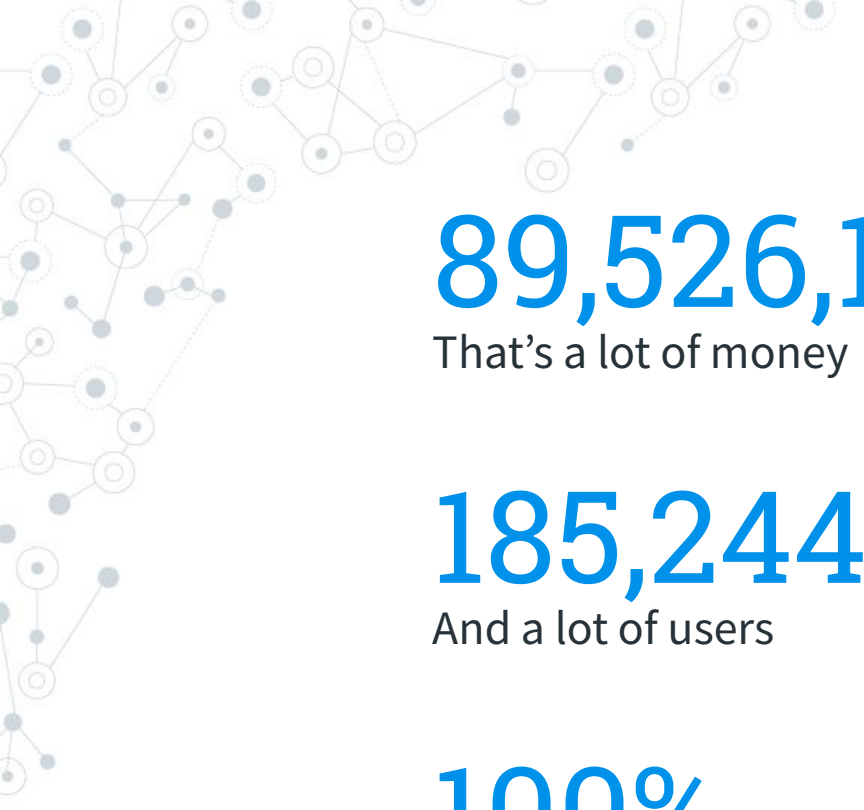
Download for free at:

<https://www.fontsquirrel.com/fonts/roboto-slab>

<https://www.fontsquirrel.com/fonts/source-sans-pro>

*You don't need to keep this slide in your presentation. It's only here to serve you as a design guide if you need to create new slides or download the fonts to edit the presentation in PowerPoint®*



A decorative network diagram in the top-left corner, featuring a complex web of interconnected nodes and lines, with some nodes highlighted in blue.

# 89,526,124\$

That's a lot of money

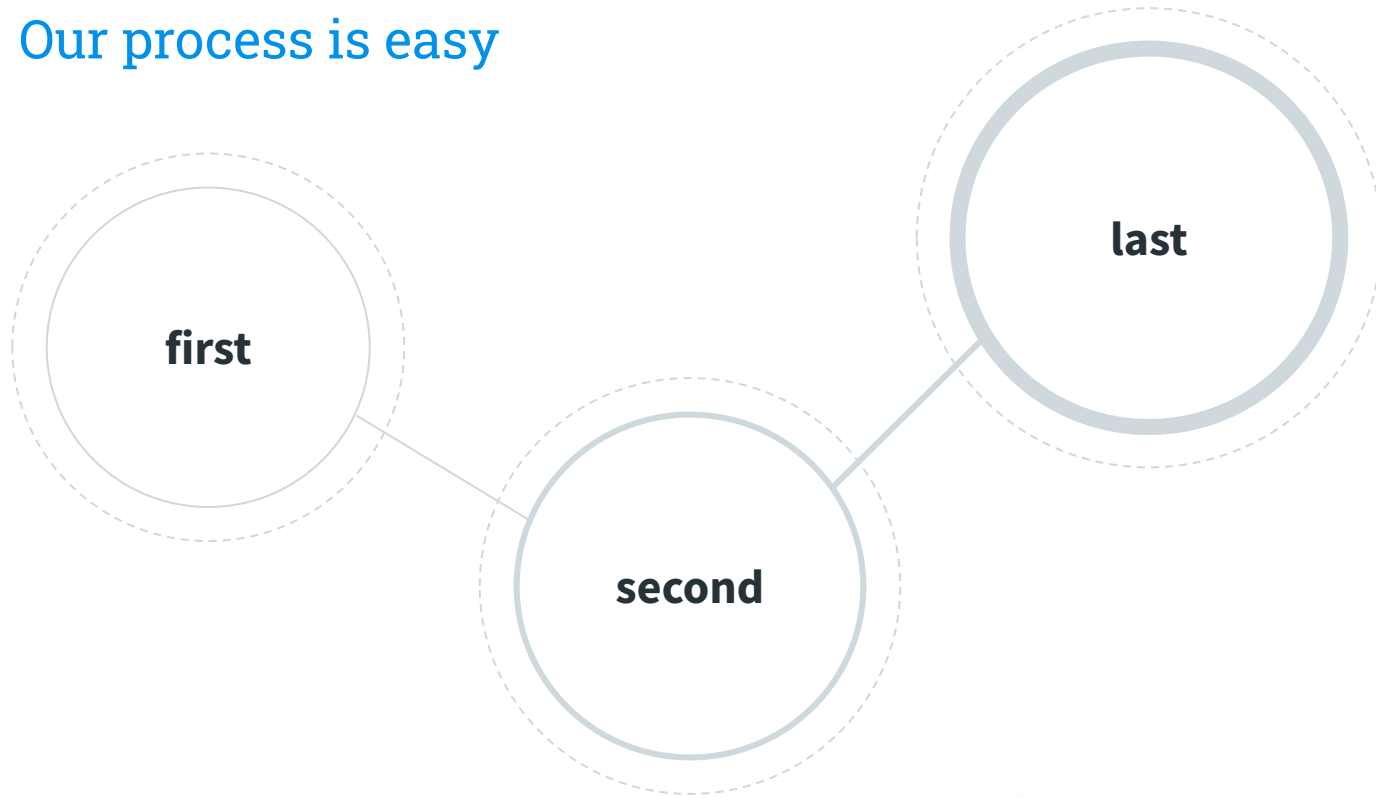
# 185,244 users

And a lot of users

# 100%

Total success!

Our process is easy



# Let's review some concepts



## Yellow

Is the color of gold, butter and ripe lemons. In the spectrum of visible light, yellow is found between green and orange.



## Yellow

Is the color of gold, butter and ripe lemons. In the spectrum of visible light, yellow is found between green and orange.



## Blue

Is the colour of the clear sky and the deep sea. It is located between violet and green on the optical spectrum.



## Blue

Is the colour of the clear sky and the deep sea. It is located between violet and green on the optical spectrum.



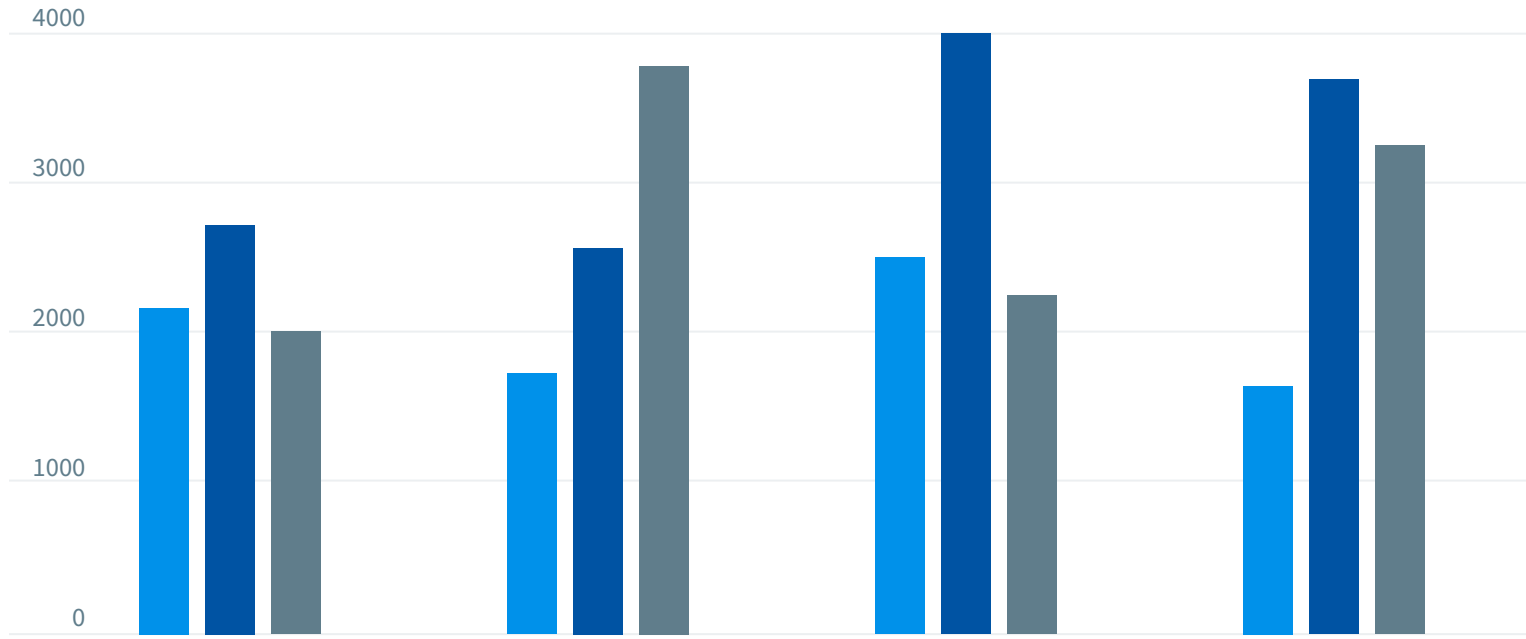
## Red

Is the color of blood, and because of this it has historically been associated with sacrifice, danger and courage.



## Red

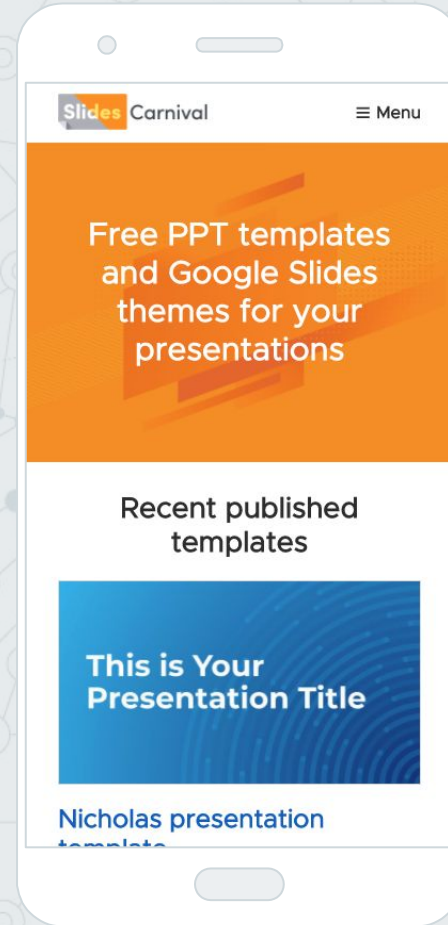
Is the color of blood, and because of this it has historically been associated with sacrifice, danger and courage.



You can insert graphs from Excel or Google Sheets

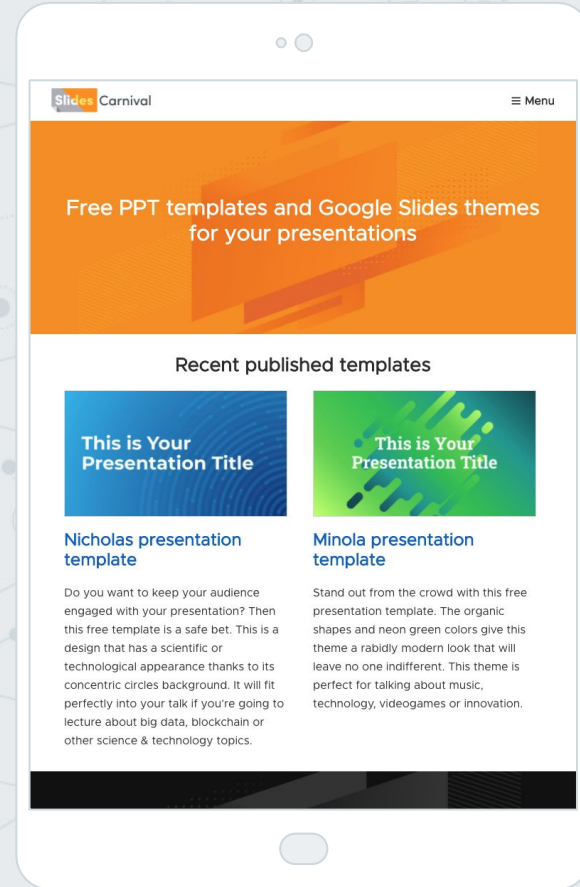
## Mobile project

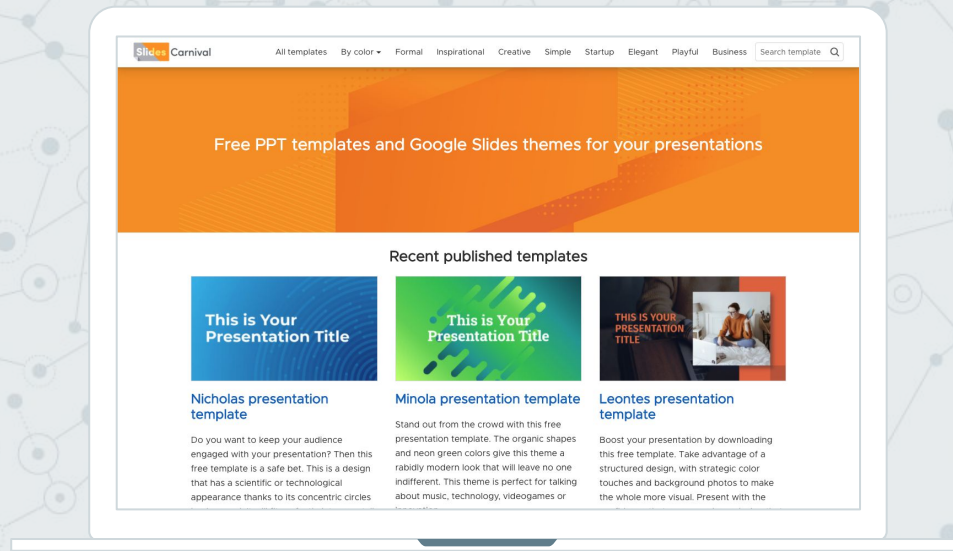
Show and explain your web, app or software projects using these gadget templates.



# Tablet project

Show and explain your web, app or software projects using these gadget templates.





## Desktop project

Show and explain your web, app or software projects using these gadget templates.



## Credits

Special thanks to all the people who made and released these awesome resources for free:

- ◎ Presentation template by SlidesCarnival
- ◎ Photographs by Unsplash



A decorative network diagram in the top-left corner, featuring a complex web of interconnected nodes and lines, with some nodes highlighted in blue and others in grey.

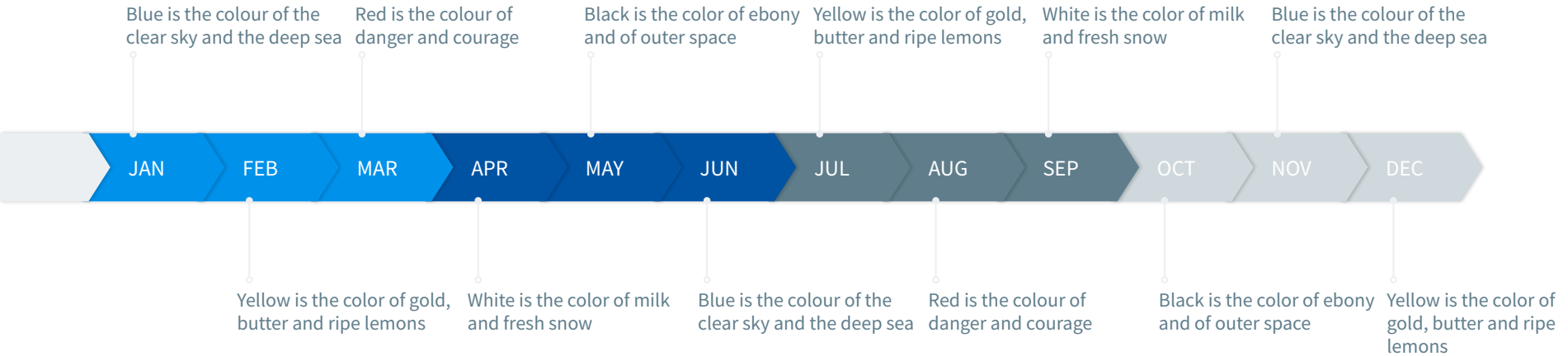
2.

## **Extra Resources**

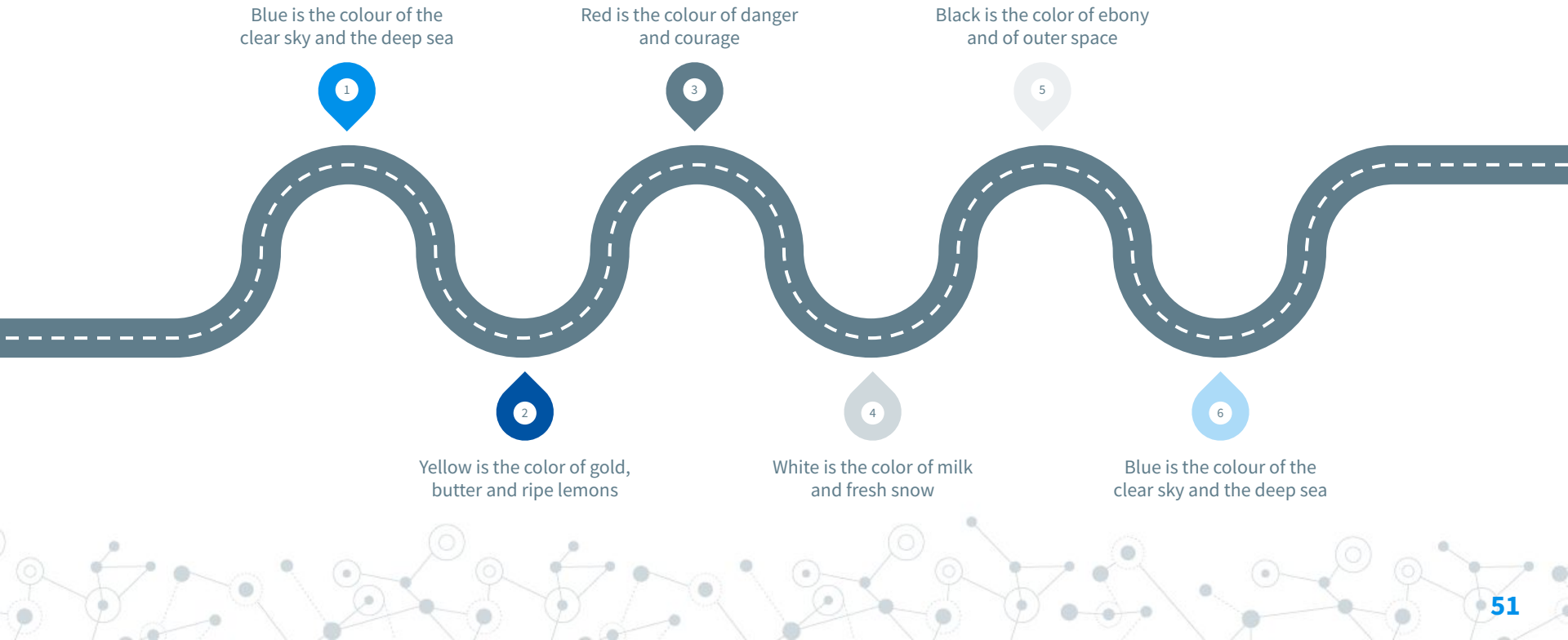
For Business Plans, Marketing Plans,  
Project Proposals, Lessons, etc

A decorative network diagram in the bottom-right corner, featuring a complex web of interconnected nodes and lines, with some nodes highlighted in blue and others in grey.

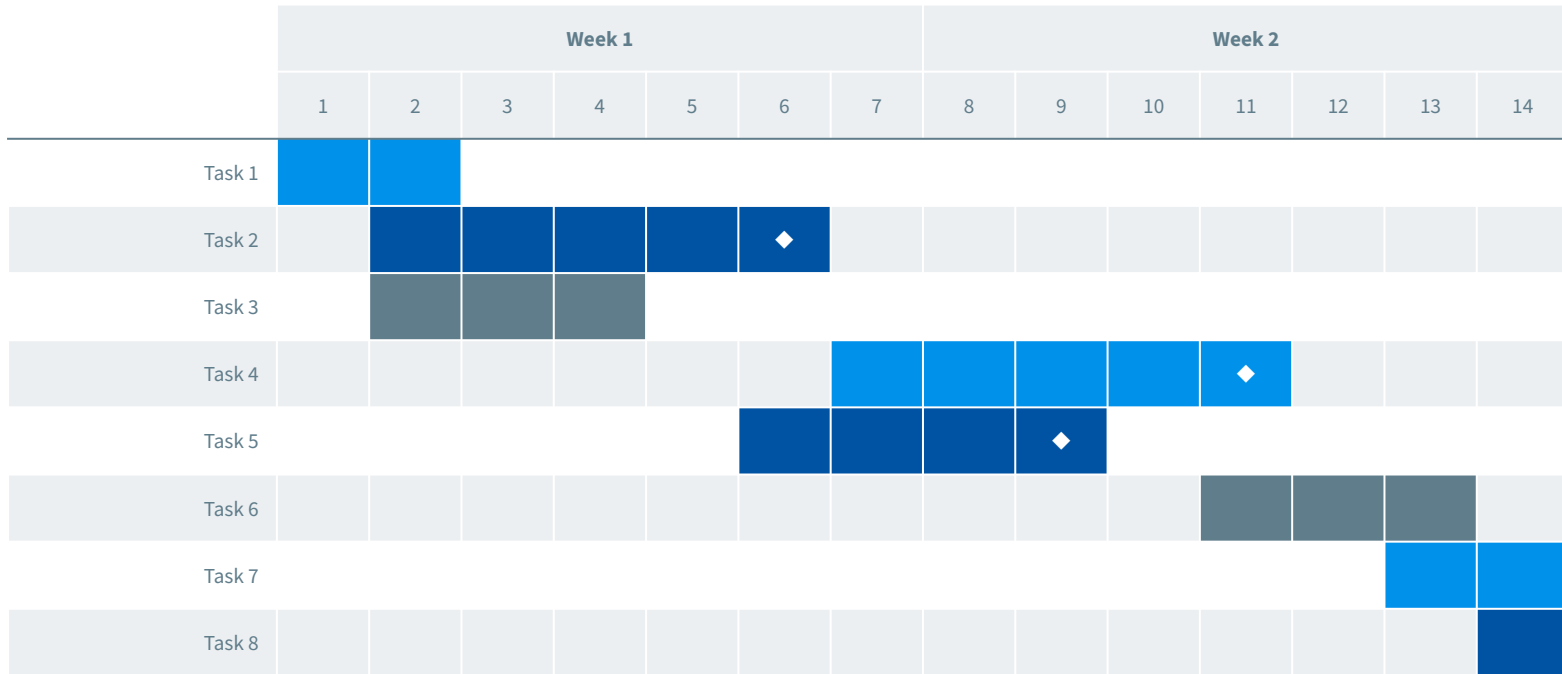
# Timeline



# Roadmap



# Gantt chart



# SWOT Analysis

## **STRENGTHS**

Blue is the colour of the clear sky and the deep sea

S

## **WEAKNESSES**

Yellow is the color of gold, butter and ripe lemons

W

O

Black is the color of ebony and of outer space

## **OPPORTUNITIES**

T

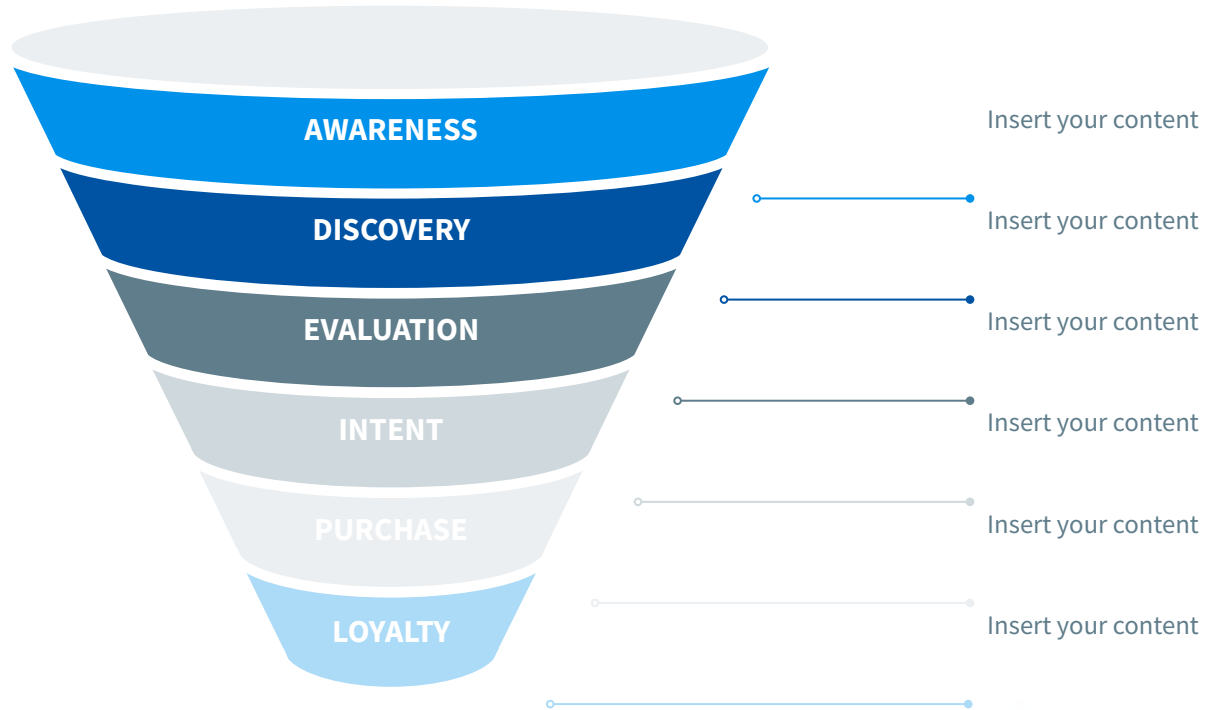
White is the color of milk and fresh snow

## **THREATS**

# Business Model Canvas

|                                                                                                                                   |                                                                                                                                   |                                                                                                                                         |                                                                                                                                             |                                                                                                                                        |
|-----------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| <b>Key Partners</b><br>Insert your content<br>   | <b>Key Activities</b><br>Insert your content<br> | <b>Value Propositions</b><br>Insert your content<br> | <b>Customer Relationships</b><br>Insert your content<br> | <b>Customer Segments</b><br>Insert your content<br> |
|                                                                                                                                   | <b>Key Resources</b><br>Insert your content<br>  |                                                                                                                                         | <b>Channels</b><br>Insert your content<br>               |                                                                                                                                        |
| <b>Cost Structure</b><br>Insert your content<br> |                                                                                                                                   |                                                                                                                                         | <b>Revenue Streams</b><br>Insert your content<br>        |                                                                                                                                        |

# Funnel



# Team Presentation



**Imani Jackson**

JOB TITLE

Blue is the colour of the clear  
sky and the deep sea



**Marcos Galán**

JOB TITLE

Blue is the colour of the clear  
sky and the deep sea



**Ixchel Valdía**

JOB TITLE

Blue is the colour of the clear  
sky and the deep sea



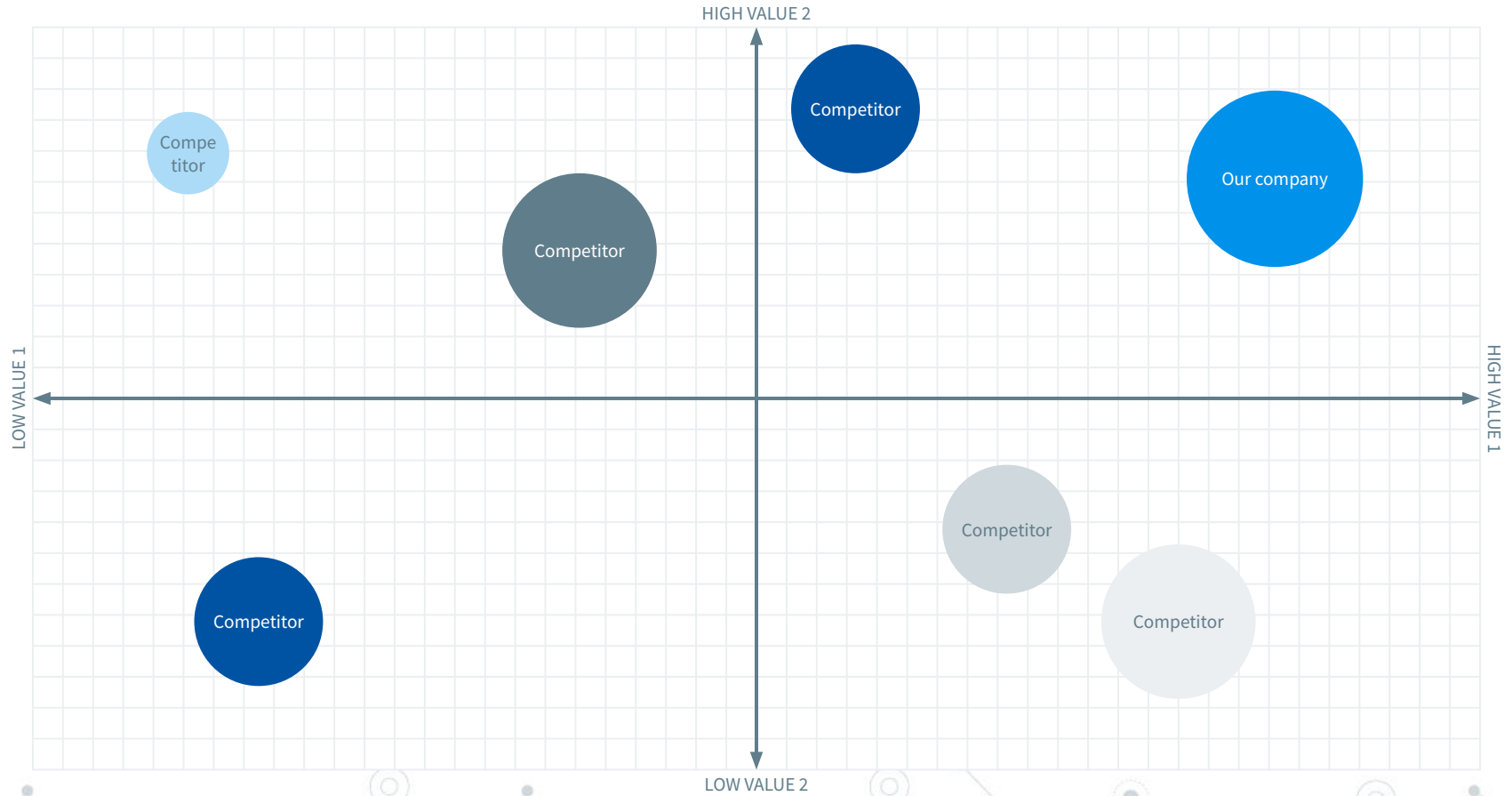
**Nils Årud**

JOB TITLE

Blue is the colour of the clear  
sky and the deep sea



## Competitor Matrix



# Weekly Planner

|               | SUNDAY      | MONDAY      | TUESDAY     | WEDNESDAY   | THURSDAY    | FRIDAY      | SATURDAY    |
|---------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|
| 09:00 - 09:45 | Task        | Task        | Task        | Task        | Task        | Task        | Task        |
| 10:00 - 10:45 | Task        | Task        | Task        | Task        | Task        | Task        | Task        |
| 11:00 - 11:45 | Task        | Task        | Task        | Task        | Task        | Task        | Task        |
| 12:00 - 13:15 | ✓ Free time | ✓ Free time | ✓ Free time | ✓ Free time | ✓ Free time | ✓ Free time | ✓ Free time |
| 13:30 - 14:15 | Task        | Task        | Task        | Task        | Task        | Task        | Task        |
| 14:30 - 15:15 | Task        | Task        | Task        | Task        | Task        | Task        | Task        |
| 15:30 - 16:15 | Task        | Task        | Task        | Task        | Task        | Task        | Task        |



## SlidesCarnival icons are editable shapes.

This means that you can:

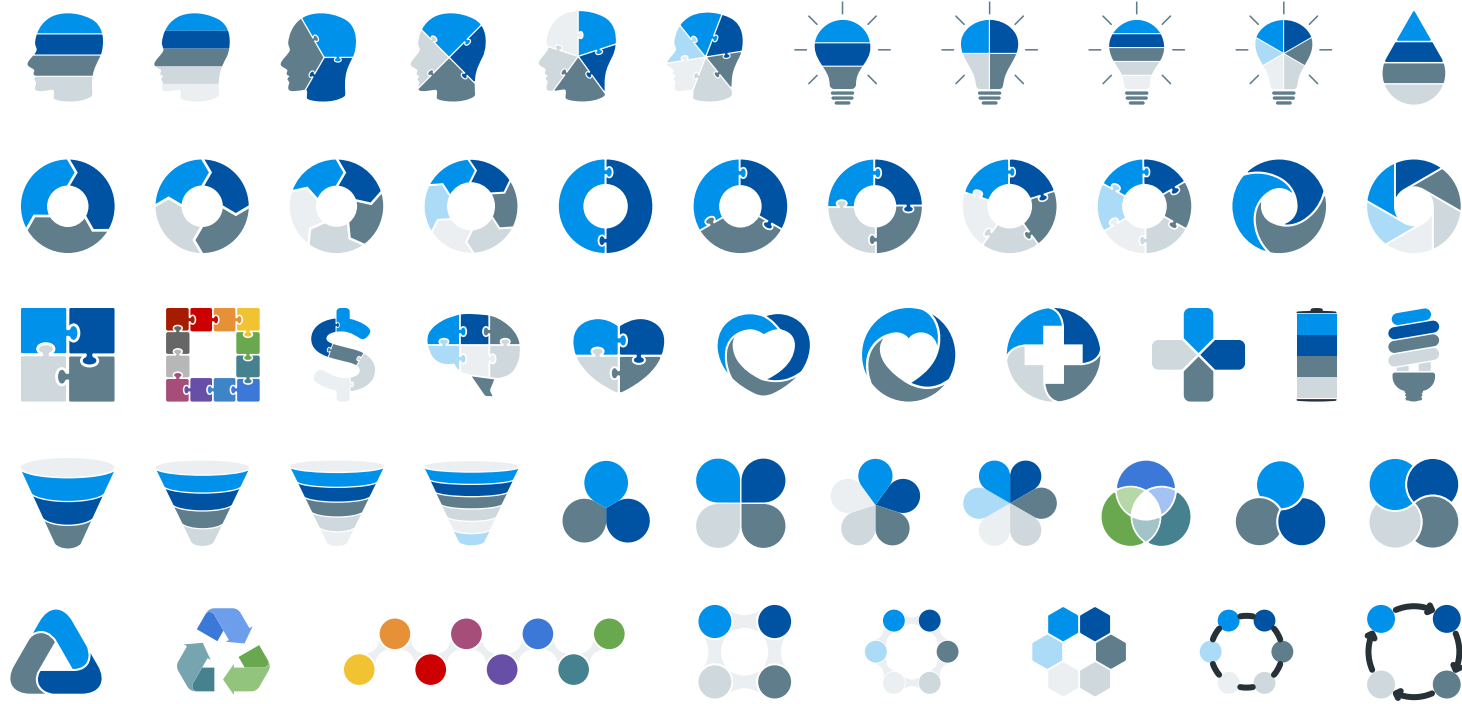
- Resize them without losing quality.
- Change line color, width and style.

Isn't that nice? :)

Examples:



# Diagrams and infographics



**You can also use any emoji as an icon!**

And of course it resizes without losing quality.

How? Follow Google instructions <https://twitter.com/googledocs/status/730087240156643328>



and many more...



**Free templates for all your presentation needs**



For PowerPoint and  
Google Slides



100% free for personal  
or commercial use



Ready to use,  
professional and  
customizable



Blow your audience  
away with attractive  
visuals